

DK LEARNING

GRADE 11

Software Engineering & Databases

Level: **O/L Advanced** • Purpose: Learn before you practise

[← BACK TO LEARNING WEBSITE](#)

In this guide

1. Software Development Life Cycle
2. Relational Databases
3. SQL Basics
4. Testing Software
5. Version Control and Teamwork
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 11 activity on the website.

Learning goals

- Describe the software development life cycle
- Design tables with primary keys
- Write basic SQL queries
- Create test cases and use version control

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: Plan your database on paper first. Fixing a design on paper is far cheaper than fixing it later.

1 Software Development Life Cycle

Software is built in stages: requirements, design, implementation (coding), testing, deployment and maintenance. Each stage produces something that guides the next.

Skipping early stages often means expensive changes later.

Key points

- Requirements first
- Design before coding
- Test before release
- Maintain after release

Worked example

Library app:
Requirements: borrow and return books
Design: tables and screens
Code -> Test -> Deploy -> Maintain

Your turn

Write requirements for a school canteen ordering app.

Check yourself

Explain each term in your own words:

SDLC

Requirement

My notes

2 Relational Databases

A relational database stores data in tables. A primary key uniquely identifies a record, while relationships connect related tables using a foreign key. Good design avoids repeating the same data.

Example: Student table and Subject table linked by a marks table.

Key points

- Tables with fields and records
- Primary key = unique ID
- Foreign key links tables
- Avoid duplicate data

Worked example

Student(StudentID PK, Name)
Subject(SubjectID PK, Title)
Mark(StudentID FK, SubjectID FK, Score)

Your turn

Design Student and Subject tables with suitable keys.

Check yourself

Explain each term in your own words:

Primary key

Foreign key

My notes

3 SQL Basics

SQL is used to work with database data. SELECT reads records; INSERT adds data; UPDATE changes data; DELETE removes data. WHERE filters rows, and ORDER BY sorts them.

Always use WHERE with UPDATE and DELETE, or you will change every row!

Key points

- SELECT reads
- INSERT adds
- UPDATE changes
- DELETE removes

Worked example

```
SELECT Name, Score
FROM Mark
WHERE Score > 50
ORDER BY Score DESC;
```

Your turn

Write a SELECT query to find students with marks above 50.

Check yourself

Explain each term in your own words:

SQL

Query

My notes

4 Testing Software

Testing checks whether software behaves as expected. Create test cases for valid, invalid and boundary inputs. Record expected and actual results. Unit testing checks small parts; system testing checks the whole.

Fix bugs and retest to confirm.

Key points

- Valid, invalid, boundary
- Expected vs actual
- Unit and system testing
- Retest after fixes

Worked example

Test: age field (allowed 5-18)

5 -> accept

18 -> accept

4 -> reject

19 -> reject

Your turn

Create test cases for a username field with 3 to 12 characters.

Check yourself

Explain each term in your own words:

Test case

Unit test

My notes

5 Version Control and Teamwork

Version control records code changes so developers can compare and restore versions. Tools such as Git track who changed what and when. Teams use branches to work on features safely.

Write clear commit messages such as "Fix login error message".

Key points

- Tracks every change
- Allows going back
- Branches for features
- Clear messages

Worked example

```
git add .  
git commit -m "Add search page"
```

Your turn

Explain how version control helps two students working on one project.

Check yourself

Explain each term in your own words:

Commit

Branch

My notes



Practice Questions

Part A - Choose the best answer

1. Which stage comes first?
 - A) Requirements
 - B) Maintenance
 - C) Deployment
2. A primary key...
 - A) uniquely identifies a record
 - B) repeats data
 - C) is a password
3. Which SQL command reads data?
 - A) SELECT
 - B) DELETE
 - C) UPDATE
4. Boundary test for 5-18 includes...
 - A) 5 and 18
 - B) only 10
 - C) only 100
5. Version control helps to...
 - A) restore earlier versions
 - B) print pages
 - C) speed the CPU
6. A foreign key...
 - A) links to another table
 - B) is always text
 - C) deletes data

Part B - Short answers

1. What does SQL INSERT do?

2. Why test before release?

3. What is a table field?

4. What is a commit?



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. Adds new data.
2. To find and fix bugs early.
3. A column holding one kind of data.
4. A saved snapshot of changes in version control.

Mini glossary

SDLC	Stages of making software
Requirement	Something the software must do
Primary key	A field that uniquely identifies each record
Foreign key	A field that links to a primary key in another table
SQL	A language for databases
Query	A request for data
Test case	An input with an expected result
Unit test	A test of a small part of a program
Commit	A saved snapshot of changes
Branch	A separate line of development



Activity Preparation

Solve database and software-testing questions and write pseudocode for a search algorithm. Then open the Grade 11 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 11 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 11 ACTIVITY**

← **EXIT PDF / BACK TO LEARNING**